

Additional Examples and Solutions on Debouncing

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Example 1 on Debouncing

A TV remote controller is to be designed with the channel selection buttons to be debounced. It was observed that the switches exhibit bounce times well under 10 ms. A **Si diode** is used across the second resistor to make the duty cycle 50%. **Design** a circuit using the 74HC14 Schmitt trigger IC along with the resistance and capacitance. The worst-case V_{th} of 74HC14 for a signal going low is 2.3 V, and that of when going high is 0.95 V. Consider that the CMOS device leakage current is 5 μA and the gate's best-case switching point is of the order of 0.5 V. **Compute** the hysteresis voltage.

Solution to Example 1 on Debouncing

During Discharging:

$$v_c = v_{th,L} = V_{final} e^{-t/R_2 C}$$

Rearranging the above equation, $R_2 = -\frac{t}{C \ln_e(\frac{V_{th,L}}{V_{final}})}$

Here, V_{final} = Final voltage before discharging = $V_{CC} - V_D = 5 - 0.7 = 4.3$ V

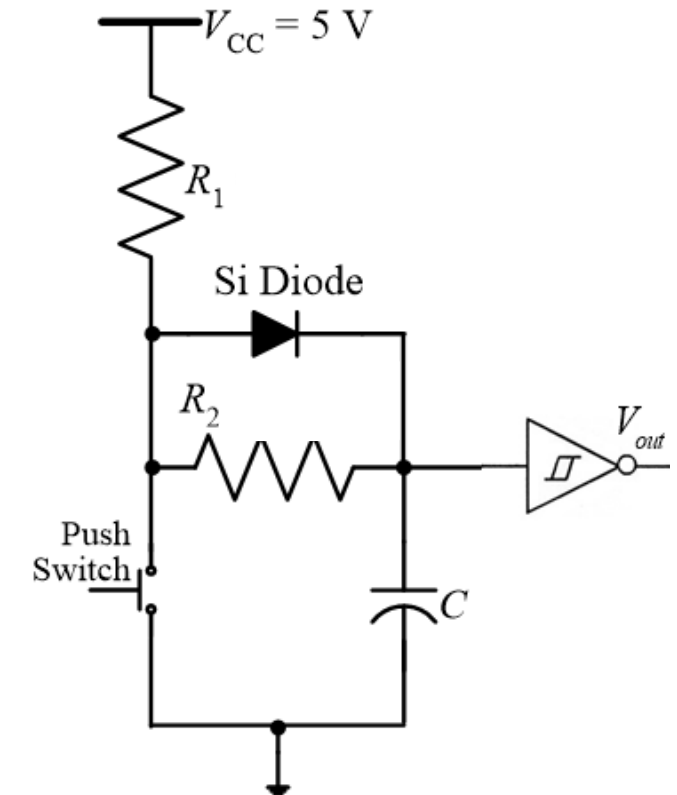
$V_{th,L} = 0.95$ V, $t = 10$ ms = 10×10^{-3} s

Assuming $C = 1$ μ F = 1×10^{-6} F

$$R_2 = \frac{-10 \times 10^{-3}}{1 \times 10^{-6} \ln \frac{0.95}{4.3}} = 6.6 \text{ k}\Omega \cong 6.8 \text{ k}\Omega$$

Assuming the Si diode's forward voltage drop, $V_D = 0.7$ V.

(6.8 k Ω is selected as this is the nearest value available with 5% tolerance)



Solution to Example 1 on Debouncing (Contd...)

Voltage drop due to leakage current of $5 \mu\text{A} = I_{\text{leakage}} R_2 = 5 \times 10^{-6} \times 6.8 \times 10^3 = 0.034 \text{ V}$ which is well below gate's best-case switching point of 0.5 V . Therefore, $C = 1 \mu\text{F}$ and $R_2 = 6.8 \text{ k}\Omega$.

During Charging:

$$v_c = v_{th,H} = V_{final} (1 - e^{-t/R_1 C}).$$

Rearranging the equation, $R_1 = -\frac{t}{C \ln_e \left(1 - \frac{V_{th,H}}{V_{final}} \right)}$

$$R_1 = \frac{-10 \times 10^{-3}}{1 \times 10^{-6} \ln \left(1 - \frac{2.3}{5 - 0.7} \right)} = 13.064 \text{ k}\Omega \cong 13 \text{ k}\Omega$$

(13 kΩ is selected as this is the nearest value available with 5% tolerance)

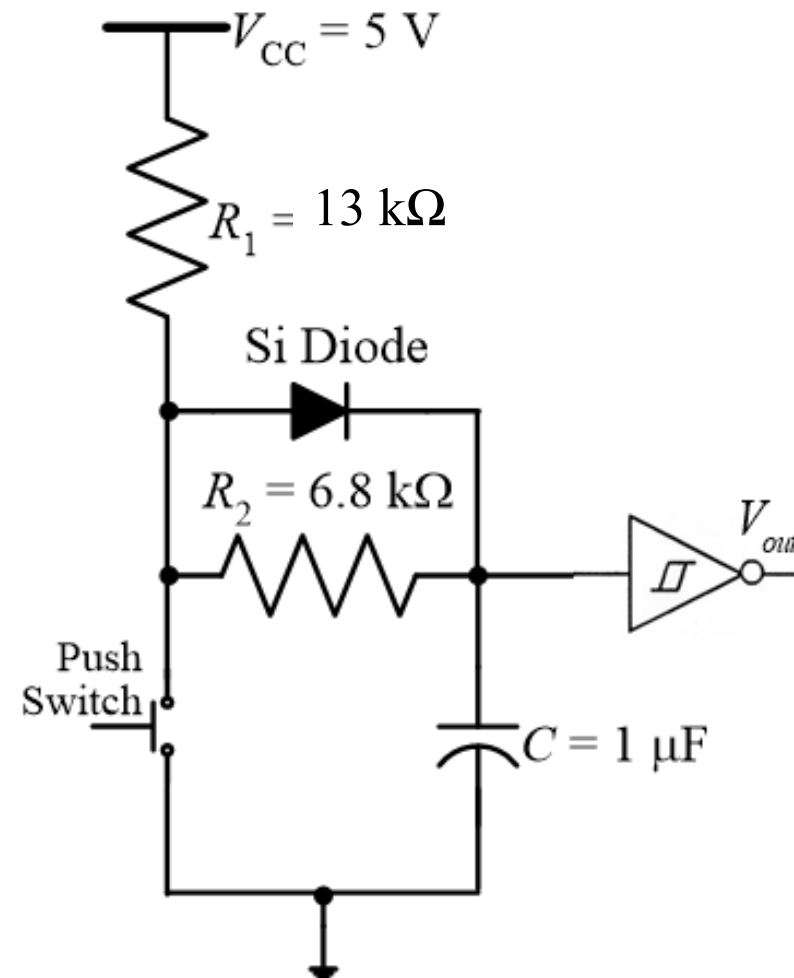
$$\text{Hysteresis voltage} = V_H = V_{th,H} - V_{th,L} = 2.3 - 0.95 = 1.35 \text{ V}$$

Solution to Example 1 on Debouncing (Contd...)

Final Design:

$R_1 = 13 \text{ k}\Omega$, $R_2 = 6.8 \text{ k}\Omega$, and $C = 1 \text{ }\mu\text{F}$.

Final Circuit:



Example 2 on Debouncing

A TV remote controller is designed with channel selection buttons to be debounced. It was observed that the switches exhibit bounce times well under 15 ms, and the duty cycle is 50%. Design a circuit using the 74HC14 Schmitt trigger IC along with resistance and capacitance. The worst-case V_{th} of this IC for a signal going low is 2.1 V, and that of when going high is 0.8 V. Also, consider that the CMOS device leakage current is 2 μ A and the gate's best-case switching point is of the order of 0.4 V. Compute the hysteresis voltage.

Solution to Example 2 on Debouncing

During Discharging:

$$v_c = v_{th,L} = V_{final} e^{-t/R_2 C}$$

Rearranging the above equation, $R_2 = -\frac{t}{C \ln_e(\frac{V_{th,L}}{V_{final}})}$

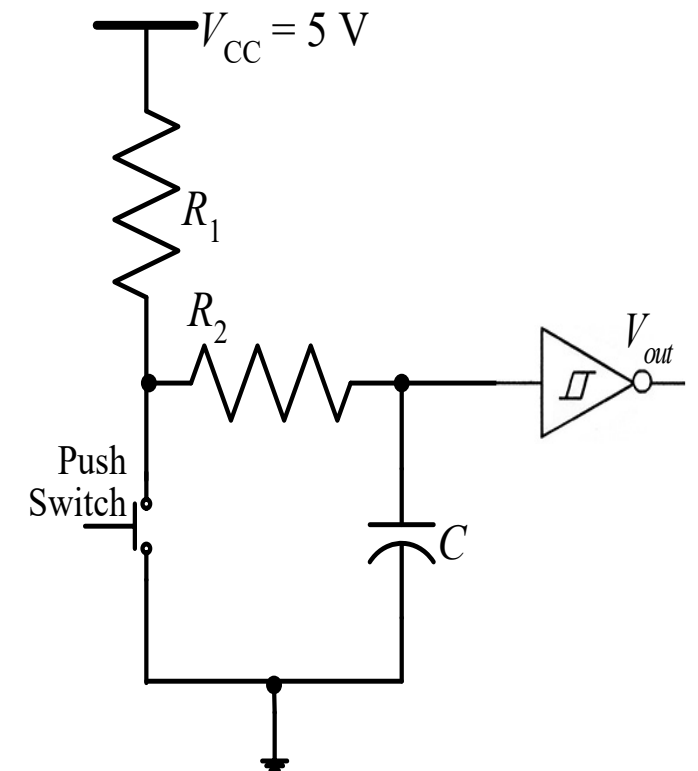
Here, V_{final} = Final voltage before discharging = $V_{CC} = 5.0 \text{ V}$

$V_{th,L} = 0.80 \text{ V}$, $t = 15 \text{ ms} = 15 \times 10^{-3} \text{ s}$

Assuming $C = 1 \text{ } \mu\text{F} = 1 \times 10^{-6} \text{ F}$

$$R_2 = \frac{-15 \times 10^{-3}}{1 \times 10^{-6} \ln \frac{0.80}{5.0}} = 8.185 \text{ k}\Omega \cong 8.2 \text{ k}\Omega$$

(8.2 k Ω is selected as this is the nearest value available with 5% tolerance)



Solution to Example 2 on Debouncing (Contd...)

Voltage drop due to leakage current of $2 \mu\text{A} = I_{\text{leakage}} R_2 = 2 \times 10^{-6} \times 8.2 \times 10^3 = 0.0164 \text{ V}$ which is well below gate's best-case switching point of 0.4 V . Therefore, $C = 1 \mu\text{F}$ and $R_2 = 8.2 \text{ k}\Omega$.

During Charging:

$$v_c = v_{th,H} = V_{final} \left(1 - e^{-t/(R_1+R_2)C} \right).$$

$$\text{Rearranging the equation, } R_1 + R_2 = -\frac{t}{C \ln_e \left(1 - \frac{V_{th,H}}{V_{final}} \right)} = \frac{-15 \times 10^{-3}}{1 \times 10^{-6} \ln \left(1 - \frac{2.1}{5} \right)} = 27.537 \text{ k}\Omega$$

$$\text{Hence, } R_1 = 27.537 - 8.2 = 19.337 \text{ k}\Omega \cong 20 \text{ k}\Omega$$

(20 k Ω is selected as this is the nearest value available with 5% tolerance)

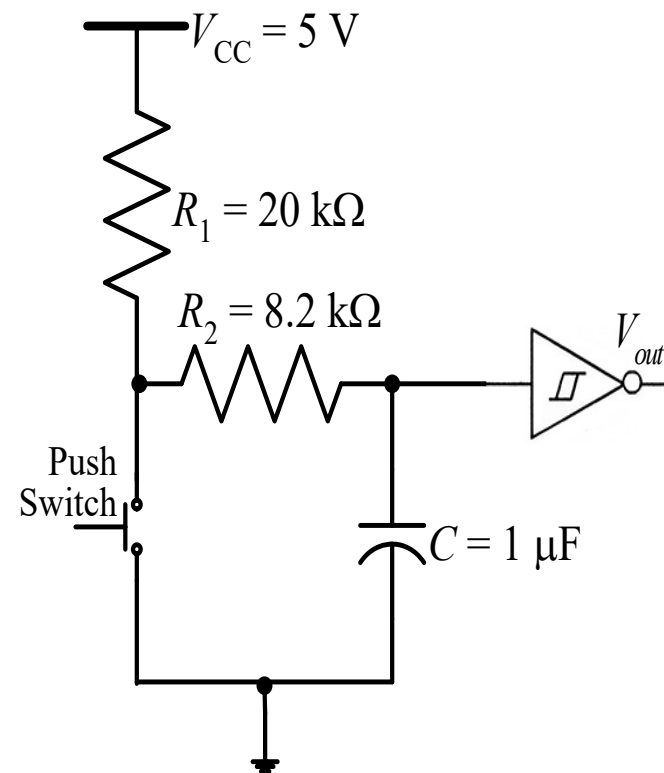
$$\text{Hysteresis voltage} = V_H = V_{th,H} - V_{th,L} = 2.1 - 0.80 = 1.30 \text{ V}$$

Solution to Example 2 on Debouncing (Contd...)

Final Design:

$R_1 = 20 \text{ k}\Omega$, $R_2 = 8.2 \text{ k}\Omega$, and $C = 1 \text{ }\mu\text{F}$.

Final Circuit:



List of nominal resistance values: 1Ω to 10 MΩ

Resistance [Ω]			IEC Code	
E6	E12	E24	3 digits J	4 digits D, F
1	1	1	1R0	1R00
		1.1	1R1	1R10
	1.2	1.2	1R2	1R20
		1.3	1R3	1R30
1.5	1.5	1.5	1R5	1R50
		1.6	1R6	1R60
	1.8	1.8	1R8	1R80
		2	2R0	2R00
2.2	2.2	2.2	2R2	2R20
		2.4	2R4	2R40
	2.7	2.7	2R7	2R70
		3	3R0	3R00
3.3	3.3	3.3	3R3	3R30
		3.6	3R6	3R60
	3.9	3.9	3R9	3R90
		4.3	4R3	4R30
4.7	4.7	4.7	4R7	4R70
		5.1	5R1	5R10
	5.6	5.6	5R6	5R60
		6.2	6R2	6R20
6.8	6.8	6.8	6R8	6R80
		7.5	7R5	7R50
	8.2	8.2	8R2	8R20
		9.1	9R1	9R10

Resistance [Ω]			IEC Code	
E6	E12	E24	3 digits J	4 digits D, F
220	220	220	221	2200
		240	241	2400
	270	270	271	2700
		300	301	3000
330	330	330	331	3300
		360	361	3600
	390	390	391	3900
		430	431	4300
470	470	470	471	4700
		510	511	5100
	560	560	561	5600
		620	621	6200
680	680	680	681	6800
		750	751	7500
	820	820	821	8200
		910	911	9100
1 k	1 k	1 k	102	1001
		1.1 k	112	1101
	1.2 k	1.2 k	122	1201
		1.3 k	132	1301
1.5 k	1.5 k	1.5 k	152	1501
		1.6 k	162	1601
	1.8 k	1.8 k	182	1801
		2 k	202	2001

Resistance [Ω]			IEC Code	
E6	E12	E24	3 digits J	4 digits D, F
47 k	47 k	47 k	473	4702
		51 k	513	5102
	56 k	56 k	563	5602
		62 k	623	6202
68 k	68 k	68 k	683	6802
		75 k	753	7502
	82 k	82 k	823	8202
		91 k	913	9102
100 k	100 k	100 k	104	1003
		110 k	114	1103
	120 k	120 k	124	1203
		130 k	134	1303
150 k	150 k	150 k	154	1503
		160 k	164	1603
	180 k	180 k	184	1803
		200 k	204	2003
220 k	220 k	220 k	224	2203
		240 k	244	2403
	270 k	270 k	274	2703
		300 k	304	3003
330 k	330 k	330 k	334	3303
		360 k	364	3603
	390 k	390 k	394	3903
		430 k	434	4303

- Tolerance D: ±0.5%, F: ±1%, J: ±5%

List of nominal resistance values: 1Ω to 10 MΩ

10	10	10	100	10R0
		11	110	11R0
	12	12	120	12R0
		13	130	13R0
15	15	15	150	15R0
		16	160	16R0
	18	18	180	18R0
		20	200	20R0
22	22	22	220	22R0
		24	240	24R0
	27	27	270	27R0
		30	300	30R0
33	33	33	330	33R0
		36	360	36R0
	39	39	390	39R0
		43	430	43R0
47	47	47	470	47R0
		51	510	51R0
	56	56	560	56R0
		62	620	62R0
68	68	68	680	68R0
		75	750	75R0
	82	82	820	82R0
		91	910	91R0
100	100	100	101	1000
		110	111	1100
	120	120	121	1200
		130	131	1300
150	150	150	151	1500
		160	161	1600
	180	180	181	1800
		200	201	2000

2.2 k	2.2 k	2.2 k	222	2201
		2.4 k	242	2401
	2.7 k	2.7 k	272	2701
		3 k	302	3001
3.3 k	3.3 k	3.3 k	332	3301
		3.6 k	362	3601
	3.9 k	3.9 k	392	3901
		4.3 k	432	4301
4.7 k	4.7 k	4.7 k	472	4701
		5.1 k	512	5101
	5.6 k	5.6 k	562	5601
		6.2 k	622	6201
6.8 k	6.8 k	6.8 k	682	6801
		7.5 k	752	7501
	8.2 k	8.2 k	822	8201
		9.1 k	912	9101
10 k	10 k	10 k	103	1002
		11 k	113	1102
	12 k	12 k	123	1202
		13 k	133	1302
15 k	15 k	15 k	153	1502
		16 k	163	1602
	18 k	18 k	183	1802
		20 k	203	2002
22 k	22 k	22 k	223	2202
		24 k	243	2402
	27 k	27 k	273	2702
		30 k	303	3002
33 k	33 k	33 k	333	3302
		36 k	363	3602
	39 k	39 k	393	3902
		43 k	433	4302

470 k	470 k	470 k	474	4703
		510 k	514	5103
	560 k	560 k	564	5603
		620 k	624	6203
680 k	680 k	680 k	684	6803
		750 k	754	7503
	820 k	820 k	824	8203
		910 k	914	9103
1 M	1 M	1 M	105	1004
		1.1 M	115	1104
	1.2 M	1.2 M	125	1204
		1.3 M	135	1304
1.5 M	1.5 M	1.5 M	155	1504
		1.6 M	165	1604
	1.8 M	1.8 M	185	1804
		2 M	205	2004
2.2 M	2.2 M	2.2 M	225	2204
		2.4 M	245	2404
	2.7 M	2.7 M	275	2704
		3 M	305	3004
3.3 M	3.3 M	3.3 M	335	3304
		3.6 M	365	3604
	3.9 M	3.9 M	395	3904
		4.3 M	435	4304
4.7 M	4.7 M	4.7 M	475	4704
		5.1 M	515	5104
	5.6 M	5.6 M	565	5604
		6.2 M	625	6204
6.8 M	6.8 M	6.8 M	685	6804
		7.5 M	755	7504
	8.2 M	8.2 M	825	8204
		9.1 M	915	9104
10 M	10 M	10 M	106	1005

- Tolerance D: ±0.5%, F: ±1%, J: ±5%

Problem 1 on Debouncing

A TV remote controller is to be designed with the channel selection buttons to be debounced. It was observed that the switches exhibit bounce times well under 20 ms. **Design** a circuit using the 74HC14 Schmitt trigger IC along with the resistance and capacitance. The worst-case V_{th} of this IC for a signal going low is 2.5 V, and that of when going high is 0.8 V. Also, consider that the CMOS device leakage current is 10 μA and the gate's best-case switching point is of the order of 0.4 V. **Compute** the hysteresis voltage.